Project Gaja-Saviya

An Intelligent Multi-Modal Framework for Human-Elephant Conflict Mitigation

R26-IT-136

Individual Project Proposal Report

Component: Dynamic Safe-Routing and Trust Platform

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Human-Elephant Conflict (HEC) in Sri Lanka leaves farmers not only facing physical danger from elephant intrusions but also uncertainty about how to respond safely once an alert is received. Current alert systems, where they exist, provide only a binary notification with no situational guidance — no indication of where the elephant is, which paths remain safe, or how much time a farmer has to reach safety. This research proposes the Dynamic Safe-Routing and Trust Platform, the user-facing application component of Project Gaja-Saviya, designed to translate validated threat data from the Cross-Modal Sensor Fusion engine into actionable, real-time safety guidance for farmers. The component is developed as a cross-platform mobile application using React Native and visualizes active threat zones on an OpenStreetMap-based interface. Upon receiving a validated threat alert, the application dynamically computes safe evacuation routes using a modified Dijkstra algorithm, where edge weights in the road network graph are adjusted in real time based on threat proximity and severity. The platform also incorporates a community-sourced elephant sighting reporting feature, governed by a trust scoring mechanism that validates user-submitted reports against sensor data and historical reporting accuracy before incorporating them into the threat map. This combination of IoT-validated sensor alerts and trust-filtered community intelligence creates a comprehensive, reliable, and actionable safety platform for HEC-affected communities.

Keywords: Safe Routing, HEC, Mobile Application, Dijkstra Algorithm, Trust Scoring, OpenStreetMap, React Native, Community Reporting

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
IoT	Internet of Things
HEC	Human-Elephant Conflict
GPS	Global Positioning System
GIS	Geographic Information System
API	Application Programming Interface
OSM	OpenStreetMap
REST	Representational State Transfer
UI	User Interface
UX	User Experience
ML	Machine Learning
FCM	Firebase Cloud Messaging
HTTPS	Hypertext Transfer Protocol Secure
JSON	JavaScript Object Notation

1. INTRODUCTION

1.1 Background and Context

When an elephant breaches a farm perimeter, the time between detection and farmer response is critical. Delays in receiving alerts, uncertainty about the elephant's location, and lack of knowledge about safe escape routes can turn a manageable situation into a life-threatening one. Despite advances in detection technology, the practical translation of threat information into farmer safety guidance has received minimal attention in HEC research. Farmers often rely on word-of-mouth or mobile phone calls — slow, informal, and unreliable mechanisms — to warn each other about nearby elephant activity.

The Dynamic Safe-Routing and Trust Platform addresses the final mile of the HEC alert chain: delivering actionable safety guidance directly to the farmer's mobile device the moment a threat is validated. By combining real-time threat zone visualization, algorithmic safe route computation, and community-sourced elephant sighting reports — all subject to trust-based filtering — this component transforms raw threat data into an intelligent safety companion.

The platform is developed using React Native to ensure cross-platform compatibility across Android and iOS devices common among Sri Lankan farmers. Map data from OpenStreetMap provides a free, locally accurate road and path network. The Dijkstra algorithm, a well-established shortest-path method, is adapted to compute dynamically weighted safe routes that avoid zones flagged by the sensor fusion engine. A trust scoring mechanism ensures that community reports do not introduce false information into the threat map without corroboration.

1.2 Literature Review

Location-aware mobile applications for emergency guidance have been extensively studied in disaster response contexts. Systems such as the FEMA app and various earthquake evacuation routing tools demonstrate the feasibility of real-time route recommendation based on dynamic hazard maps. Graph-based routing algorithms, particularly Dijkstra and A* (Hart et al., 1968), are the foundation of most navigation applications and have been adapted for hazard-aware routing by modifying edge weights to penalize routes through danger zones.

OpenStreetMap (OSM) has emerged as a reliable, community-maintained geographic data source particularly valuable in rural and developing-world contexts where commercial map providers have sparse coverage. OSM's road network data can be accessed programmatically via the Overpass API and processed using libraries such as OSMnx (Boeing, 2017) to construct weighted graph representations suitable for algorithmic routing.

Community-sourced information systems have demonstrated significant value in wildlife monitoring and conservation, enabling large-scale spatial coverage beyond what sensor networks alone can provide. However, crowdsourced data is inherently noisy and subject to false reports. Trust scoring mechanisms — where reporters' historical accuracy influences the weight given to their submissions — have been proposed and implemented in citizen science

platforms. Research by Sheng et al. (2008) on quality control in crowdsourcing provides a theoretical foundation for the trust model applied in this component.

React Native has become the dominant framework for cross-platform mobile application development due to its performance characteristics, code reuse between iOS and Android, and extensive community support. Push notification services such as Firebase Cloud Messaging (FCM) enable reliable, low-latency delivery of alerts to mobile devices even when the application is in the background.

1.3 Research Gap

While routing algorithms and community reporting systems are well-studied individually, no published system has combined IoT-validated sensor data, real-time threat-aware route computation, and trust-filtered community reporting in a unified mobile platform specifically designed for HEC emergency guidance. Furthermore, the specific challenges of rural Sri Lankan road networks — limited road data quality, unpaved paths, and seasonal route accessibility — have not been addressed in existing routing research. The integration of continuously updated threat zones from an AI fusion engine as dynamic edge weights in a real-time routing graph represents a novel contribution. This research addresses all of these gaps.

2. OBJECTIVES

2.1 Main Objective

To design and develop a cross-platform mobile application that receives validated threat alerts from the sensor fusion engine, visualizes active threat zones on an interactive map, computes dynamically weighted safe evacuation routes, and incorporates trust-filtered community elephant sighting reports to provide comprehensive real-time safety guidance to farmers in HEC-affected areas.

2.2 Specific Objectives

1. To develop a React Native mobile application compatible with Android and iOS, providing an intuitive map-based interface for threat visualization and safe routing.
2. To integrate OpenStreetMap data for the target deployment region, construct a road network graph, and implement the Dijkstra algorithm for shortest-path routing.
3. To implement a dynamic edge-weight adjustment mechanism that increases routing cost through road segments proximal to validated threat zones based on threat severity.
4. To design and implement a community elephant sighting reporting feature with timestamp, location, and optional notes.
5. To develop a trust scoring model that evaluates user-submitted reports based on historical reporting accuracy, proximity to sensor-validated events, and recency.
6. To implement Firebase Cloud Messaging push notifications for real-time alert delivery with less than 5-second end-to-end latency from fusion output to farmer device.
7. To evaluate the application's usability through structured user testing with target farmer population representatives.

3. METHODOLOGY

This research follows the Design Science Research methodology, producing a functional mobile application prototype through iterative design, development, and user evaluation cycles.

Phase 1: Map Data Acquisition and Graph Construction

OpenStreetMap data for the target deployment region (identified HEC zones in Sri Lanka) will be downloaded using the Overpass API. The OSMnx Python library will be used to construct a weighted directed graph where nodes represent road intersections and edges represent road segments with weights representing travel time and safety score. The graph will be stored in the backend database and served to the mobile application via a REST API.

Phase 2: Mobile Application Development

The React Native application will be developed with three core screens: a live map dashboard showing current threat zones and safe routes; a community reporting screen for submitting and viewing elephant sightings; and an alert history screen. The map will be rendered using the react-native-maps library with OSM tile provider. GPS location tracking will continuously update the user's position on the map. The application will communicate with the backend via HTTPS REST APIs.

Phase 3: Dynamic Routing Algorithm

Upon receiving a validated threat alert from the fusion engine (via webhook), the backend routing service will identify all road segments within a configurable radius of the threat location (default: 500m for high severity, 200m for medium). Edge weights for these segments will be multiplied by a threat penalty factor proportional to severity (1.5x for medium, 3x for high). The modified Dijkstra algorithm will then compute the safest route from the user's current GPS location to the nearest safe zone. The computed route will be pushed to the user's device via FCM.

Phase 4: Trust Scoring System

When a user submits a community sighting report, the system will assign an initial trust score based on: proximity to a recent sensor-validated event (high boost), the reporter's historical accuracy score (weighted average of past report validation outcomes), and report recency. Reports with trust scores above a configurable threshold will be displayed on the threat map; low-trust reports will be queued for review. The trust model will be implemented as a Python service on the backend.

4. SYSTEM ARCHITECTURE

4.1 Component Position

The Dynamic Safe-Routing and Trust Platform is the downstream consumer and user-facing output of the entire Gaja-Saviya pipeline. It receives validated threat objects from the Cross-Modal Sensor Fusion API via a backend webhook, processes them into routing decisions, and delivers alerts and navigation to farmers via the mobile application.

4.2 Internal Sub-Components

- Alert Receiver Service: Backend webhook listener consuming validated threat objects from the Fusion API.
- Routing Engine: Dijkstra-based safe route computation with dynamic threat-based edge weight adjustment.
- Map Tile Server Proxy: Caches and serves OSM map tiles for offline-capable map display.
- Community Report Service: Receives, validates, and scores user-submitted elephant sighting reports.
- Trust Scoring Engine: Computes and updates trust scores for community reporters and reports.
- Push Notification Service: Dispatches FCM alerts to registered farmer devices.
- React Native Mobile App: User-facing application with map dashboard, community reporting, and alert history.

4.3 Data Flow

Fusion API -> Alert Receiver -> Routing Engine -> [Push Notification Service -> Farmer Device App] AND [Map Dashboard Update]

Farmer Device -> Community Report Service -> Trust Scoring Engine -> [Approved: Map Dashboard] OR [Queued: Review]

5. USER REQUIREMENTS

5.1 Stakeholders

- Primary Users: Farmers in HEC-affected areas who use the mobile app for alerts and safe navigation.
- Community Reporters: Farmers or community members who submit elephant sighting reports.
- System Administrators: Technical staff managing the backend services and trust scoring parameters.
- Fusion Component: Upstream provider of validated threat data consumed by this component.

5.2 Functional Requirements

8. The system shall display active elephant threat zones as highlighted polygons on an interactive map.
9. The system shall compute and display a safe evacuation route from the user's current GPS location upon receiving a high-severity alert.
10. The system shall dynamically update route edge weights based on threat proximity and severity.
11. The system shall send push notifications to registered farmer devices within 5 seconds of receiving a validated threat from the fusion engine.
12. The system shall allow users to submit community elephant sighting reports with location, timestamp, and optional description.
13. The system shall apply trust scoring to community reports and only display reports above the minimum trust threshold on the map.
14. The system shall maintain an alert history log accessible to users for reviewing past events.
15. The system shall support offline map caching for areas with unreliable internet connectivity.

5.3 Non-Functional Requirements

Requirement	Specification
Alert Latency	Push notification delivered within 5 seconds of fusion output
Route Computation	Safe route computed and displayed within 3 seconds of alert receipt
Usability	System Usability Scale score above 75 in farmer user testing
Platform Support	Compatible with Android 9+ and iOS 13+
Offline Capability	Core map and last known threat zones accessible without internet

Security	All API communications encrypted via HTTPS/TLS; user authentication required
Scalability	Support 1,000 concurrent registered users

6. COMMERCIALIZATION PLAN

6.1 Target Market

The primary target users are farming households in HEC-affected districts of Sri Lanka, including Ampara, Polonnaruwa, Anuradhapura, and Ratnapura. Secondary stakeholders include the Department of Wildlife Conservation, NGOs working in conflict mitigation, and district emergency management authorities who could use the platform's data for strategic planning.

6.2 Value Proposition

The platform transforms the HEC response from reactive and uninformed to proactive and data-guided. Farmers receive actionable safety guidance — not just an alarm — within seconds of a validated intrusion event. The community reporting feature builds a shared situational awareness that extends coverage beyond sensor networks and empowers communities to protect each other.

6.3 Pricing Strategy

Tier	Price	Features
Free	Rs. 0	Basic alert notifications and map view
Farmer Premium	Rs. 299/month	Full routing, community reporting, offline maps
DWC/NGO Institutional	Rs. 15,000/month	Multi-zone dashboard, analytics, API access

7. BUDGET AND JUSTIFICATION

Category	Amount (LKR)
Cloud Backend (Firebase + Hosting)	22,000
Mobile App Development Tools (Expo EAS)	8,500
OSM Data Processing & Hosting	5,000
User Testing Participants	6,000
UI/UX Design Tools (Figma)	4,000
Contingency	4,500
TOTAL	**50,000**

Firebase services cover hosting, real-time database, and FCM push notifications — essential for the always-on alert delivery system. Expo EAS Build enables compilation and deployment of the React Native app to both app stores. User testing costs cover travel and compensation for farmer participants in usability evaluation sessions in target HEC districts.

8. WORK BREAKDOWN STRUCTURE

Phase	Task	Duration
1	Literature Review & Proposal Preparation	Months 1-2
2	OSM Data Acquisition & Road Network Graph Construction	Months 2-3
3	React Native App Skeleton & Map Integration	Months 3-4
4	Dynamic Routing Algorithm Implementation	Months 4-5
5	Community Reporting & Trust Scoring System	Months 5-6
6	Push Notification & Fusion API Integration	Month 6
7	Cross-Component System Integration Testing	Months 6-7
8	Farmer User Testing & Usability Evaluation	Months 7-8
9	Documentation & Final Submission	Months 8-9

9. RISK ANALYSIS

Risk	Impact	Mitigation
Poor internet connectivity in rural deployment zones	High	Implement offline map caching and store-and-forward alert queue
Inaccurate OSM road data in rural areas	High	Supplement with local knowledge; allow manual path corrections via admin interface
Low trust score degrading community report quality	Medium	Calibrate trust thresholds carefully; review queued reports manually during pilot
High FCM alert latency	Medium	Implement persistent socket connection as fallback to FCM push
Farmer user adoption challenges	Medium	Conduct co-design workshops; provide multilingual (Sinhala/Tamil) interface
Push notification delivery failure on low-end devices	Low	Test on common device models; provide SMS fallback for critical alerts

10. REFERENCES

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